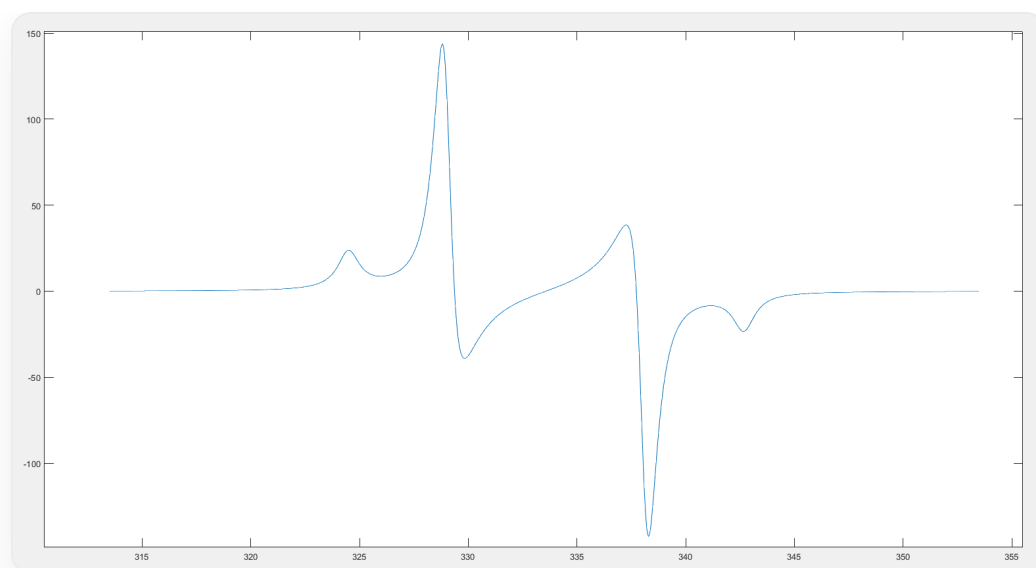


题目

一种碳中心有机双自由基在冻结甲苯中的EPR模拟谱图如下所示，试根据谱图解析指出该双自由基两个单电子之间的平均间距（以pm为单位）。



该图展示了一张EPR谱图，横坐标为磁场强度（单位为mT），纵轴为强度（单位任意），其中的数据点如下，每个括号内为一个数据点，其中第一个数据为磁场强度（单位为mT），第二个数据为峰强度（单位任意）：(3.1350000e+02 1.4867028e-04)(3.1353910e+02 1.4972312e-04)(3.1357820e+02 1.5078641e-04)

(3.1361730e+02 1.5186027e-04)(3.1365640e+02 1.5294486e-04)(3.1369550e+02 1.5404029e-04)
 (3.1373460e+02 1.5514673e-04)(3.1377370e+02 1.5626431e-04)(3.1381281e+02 1.5739318e-04)
 (3.1385191e+02 1.5853350e-04)(3.1389101e+02 1.5968540e-04)(3.1393011e+02 1.6084906e-04)
 (3.1396921e+02 1.6202461e-04)(3.1400831e+02 1.6321223e-04)(3.1404741e+02 1.6441208e-04)
 (3.1408651e+02 1.6562432e-04)(3.1412561e+02 1.6684913e-04)(3.1416471e+02 1.6808666e-04)
 (3.1420381e+02 1.6933710e-04)(3.1424291e+02 1.7060062e-04)(3.1428201e+02 1.7187740e-04)
 (3.1432111e+02 1.7316763e-04)(3.1436022e+02 1.7447149e-04)(3.1439932e+02 1.7578917e-04)
 (3.1443842e+02 1.7712087e-04)(3.1447752e+02 1.7846678e-04)(3.1451662e+02 1.7982709e-04)
 (3.1455572e+02 1.8120202e-04)(3.1459482e+02 1.8259177e-04)(3.1463392e+02 1.8399655e-04)
 (3.1467302e+02 1.8541657e-04)(3.1471212e+02 1.8685206e-04)(3.1475122e+02 1.8830323e-04)
 (3.1479032e+02 1.8977031e-04)(3.1482942e+02 1.9125353e-04)(3.1486852e+02 1.9275313e-04)
 (3.1490762e+02 1.9426934e-04)(3.1494673e+02 1.9580241e-04)(3.1498583e+02 1.9735258e-04)
 (3.1502493e+02 1.9892011e-04)(3.1506403e+02 2.0050525e-04)(3.1510313e+02 2.0210827e-04)
 (3.1514223e+02 2.0372943e-04)(3.1518133e+02 2.0536900e-04)(3.1522043e+02 2.0702726e-04)
 (3.1525953e+02 2.0870450e-04)(3.1529863e+02 2.1040099e-04)(3.1533773e+02 2.1211703e-04)
 (3.1537683e+02 2.1385292e-04)(3.1541593e+02 2.1560897e-04)(3.1545503e+02 2.1738548e-04)
 (3.1549413e+02 2.1918277e-04)(3.1553324e+02 2.2100117e-04)(3.1557234e+02 2.2284099e-04)
 (3.1561144e+02 2.2470258e-04)(3.1565054e+02 2.2658628e-04)(3.1568964e+02 2.2849244e-04)
 (3.1572874e+02 2.3042141e-04)(3.1576784e+02 2.3237355e-04)(3.1580694e+02 2.3434924e-04)
 (3.1584604e+02 2.3634885e-04)(3.1588514e+02 2.3837276e-04)(3.1592424e+02 2.4042138e-04)
 (3.1596334e+02 2.4249509e-04)(3.1600244e+02 2.4459432e-04)(3.1604154e+02 2.4671948e-04)
 (3.1608065e+02 2.4887099e-04)(3.1611975e+02 2.5104929e-04)(3.1615885e+02 2.5325482e-04)

(3.1619795e+02 2.5548804e-04)(3.1623705e+02 2.5774942e-04)(3.1627615e+02 2.6003942e-04)
 (3.1631525e+02 2.6235852e-04)(3.1635435e+02 2.6470723e-04)(3.1639345e+02 2.6708605e-04)
 (3.1643255e+02 2.6949549e-04)(3.1647165e+02 2.7193608e-04)(3.1651075e+02 2.7440835e-04)
 (3.1654985e+02 2.7691287e-04)(3.1658895e+02 2.7945018e-04)(3.1662805e+02 2.8202086e-04)
 (3.1666716e+02 2.8462551e-04)(3.1670626e+02 2.8726472e-04)(3.1674536e+02 2.8993911e-04)
 (3.1678446e+02 2.9264931e-04)(3.1682356e+02 2.9539595e-04)(3.1686266e+02 2.9817970e-04)
 (3.1690176e+02 3.0100122e-04)(3.1694086e+02 3.0386121e-04)(3.1697996e+02 3.0676037e-04)
 (3.1701906e+02 3.0969942e-04)(3.1705816e+02 3.1267910e-04)(3.1709726e+02 3.1570016e-04)
 (3.1713636e+02 3.1876338e-04)(3.1717546e+02 3.2186954e-04)(3.1721457e+02 3.2501946e-04)
 (3.1725367e+02 3.2821397e-04)(3.1729277e+02 3.3145391e-04)(3.1733187e+02 3.3474017e-04)
 (3.1737097e+02 3.3807362e-04)(3.1741007e+02 3.4145518e-04)(3.1744917e+02 3.4488580e-04)
 (3.1748827e+02 3.4836642e-04)(3.1752737e+02 3.5189805e-04)(3.1756647e+02 3.5548167e-04)
 (3.1760557e+02 3.5911834e-04)(3.1764467e+02 3.6280910e-04)(3.1768377e+02 3.6655506e-04)
 (3.1772287e+02 3.7035732e-04)(3.1776197e+02 3.7421703e-04)(3.1780108e+02 3.7813537e-04)
 (3.1784018e+02 3.8211355e-04)(3.1787928e+02 3.8615279e-04)(3.1791838e+02 3.9025438e-04)
 (3.1795748e+02 3.9441962e-04)(3.1799658e+02 3.9864985e-04)(3.1803568e+02 4.0294644e-04)
 (3.1807478e+02 4.0731082e-04)(3.1811388e+02 4.1174443e-04)(3.1815298e+02 4.1624877e-04)
 (3.1819208e+02 4.2082537e-04)(3.1823118e+02 4.2547582e-04)(3.1827028e+02 4.3020173e-04)
 (3.1830938e+02 4.3500477e-04)(3.1834848e+02 4.3988666e-04)(3.1838759e+02 4.4484916e-04)
 (3.1842669e+02 4.4989409e-04)(3.1846579e+02 4.5502333e-04)(3.1850489e+02 4.6023878e-04)
 (3.1854399e+02 4.6554244e-04)(3.1858309e+02 4.7093635e-04)(3.1862219e+02 4.7642259e-04)
 (3.1866129e+02 4.8200334e-04)(3.1870039e+02 4.8768083e-04)(3.1873949e+02 4.9345735e-04)
 (3.1877859e+02 4.9933528e-04)(3.1881769e+02 5.0531704e-04)(3.1885679e+02 5.1140516e-04)
 (3.1889589e+02 5.1760224e-04)(3.1893500e+02 5.2391094e-04)(3.1897410e+02 5.3033404e-04)
 (3.1901320e+02 5.3687439e-04)(3.1905230e+02 5.4353493e-04)(3.1909140e+02 5.5031870e-04)
 (3.1913050e+02 5.5722883e-04)(3.1916960e+02 5.6426858e-04)(3.1920870e+02 5.7144129e-04)
 (3.1924780e+02 5.7875043e-04)(3.1928690e+02 5.8619958e-04)(3.1932600e+02 5.9379243e-04)
 (3.1936510e+02 6.0153282e-04)(3.1940420e+02 6.0942471e-04)(3.1944330e+02 6.1747220e-04)
 (3.1948240e+02 6.2567954e-04)(3.1952151e+02 6.3405111e-04)(3.1956061e+02 6.4259148e-04)
 (3.1959971e+02 6.5130535e-04)(3.1963881e+02 6.6019761e-04)(3.1967791e+02 6.6927332e-04)
 (3.1971701e+02 6.7853775e-04)(3.1975611e+02 6.8799633e-04)(3.1979521e+02 6.9765472e-04)
 (3.1983431e+02 7.0751879e-04)(3.1987341e+02 7.1759462e-04)(3.1991251e+02 7.2788854e-04)
 (3.1995161e+02 7.3840712e-04)(3.1999071e+02 7.4915719e-04)(3.2002981e+02 7.6014584e-04)
 (3.2006891e+02 7.7138047e-04)(3.2010802e+02 7.8286874e-04)(3.2014712e+02 7.9461864e-04)
 (3.2018622e+02 8.0663850e-04)(3.2022532e+02 8.1893698e-04)(3.2026442e+02 8.3152310e-04)
 (3.2030352e+02 8.4440626e-04)(3.2034262e+02 8.5759627e-04)(3.2038172e+02 8.7110335e-04)
 (3.2042082e+02 8.8493817e-04)(3.2045992e+02 8.9911187e-04)(3.2049902e+02 9.1363608e-04)

(3.2053812e+02 9.2852294e-04)(3.2057722e+02 9.4378515e-04)(3.2061632e+02 9.5943598e-04)
 (3.2065543e+02 9.7548932e-04)(3.2069453e+02 9.9195969e-04)(3.2073363e+02 1.0088623e-03)
 (3.2077273e+02 1.0262131e-03)(3.2081183e+02 1.0440288e-03)(3.2085093e+02 1.0623268e-03)
 (3.2089003e+02 1.0811256e-03)(3.2092913e+02 1.1004444e-03)(3.2096823e+02 1.1203034e-03)
 (3.2100733e+02 1.1407239e-03)(3.2104643e+02 1.1617283e-03)(3.2108553e+02 1.1833400e-03)
 (3.2112463e+02 1.2055837e-03)(3.2116373e+02 1.2284854e-03)(3.2120283e+02 1.2520727e-03)
 (3.2124194e+02 1.2763743e-03)(3.2128104e+02 1.3014208e-03)(3.2132014e+02 1.3272443e-03)
 (3.2135924e+02 1.3538787e-03)(3.2139834e+02 1.3813599e-03)(3.2143744e+02 1.4097258e-03)
 (3.2147654e+02 1.4390166e-03)(3.2151564e+02 1.4692747e-03)(3.2155474e+02 1.5005451e-03)
 (3.2159384e+02 1.5328754e-03)(3.2163294e+02 1.5663163e-03)(3.2167204e+02 1.6009213e-03)
 (3.2171114e+02 1.6367476e-03)(3.2175024e+02 1.6738557e-03)(3.2178935e+02 1.7123101e-03)
 (3.2182845e+02 1.7521794e-03)(3.2186755e+02 1.7935368e-03)(3.2190665e+02 1.8364603e-03)
 (3.2194575e+02 1.8810329e-03)(3.2198485e+02 1.9273436e-03)(3.2202395e+02 1.9754874e-03)
 (3.2206305e+02 2.0255658e-03)(3.2210215e+02 2.0776876e-03)(3.2214125e+02 2.1319693e-03)
 (3.2218035e+02 2.1885359e-03)(3.2221945e+02 2.2475217e-03)(3.2225855e+02 2.3090707e-03)
 (3.2229765e+02 2.3733380e-03)(3.2233675e+02 2.4404905e-03)(3.2237586e+02 2.5107080e-03)
 (3.2241496e+02 2.5841844e-03)(3.2245406e+02 2.6611290e-03)(3.2249316e+02 2.7417679e-03)
 (3.2253226e+02 2.8263459e-03)(3.2257136e+02 2.9151276e-03)(3.2261046e+02 3.0084001e-03)
 (3.2264956e+02 3.1064747e-03)(3.2268866e+02 3.2096893e-03)(3.2272776e+02 3.3184113e-03)
 (3.2276686e+02 3.4330402e-03)(3.2280596e+02 3.5540112e-03)(3.2284506e+02 3.6817982e-03)
 (3.2288416e+02 3.8169185e-03)(3.2292326e+02 3.9599365e-03)(3.2296237e+02 4.1114688e-03)
 (3.2300147e+02 4.2721897e-03)(3.2304057e+02 4.4428363e-03)(3.2307967e+02 4.6242155e-03)
 (3.2311877e+02 4.8172106e-03)(3.2315787e+02 5.0227887e-03)(3.2319697e+02 5.2420086e-03)
 (3.2323607e+02 5.4760294e-03)(3.2327517e+02 5.7261189e-03)(3.2331427e+02 5.9936631e-03)
 (3.2335337e+02 6.2801747e-03)(3.2339247e+02 6.5873018e-03)(3.2343157e+02 6.9168350e-03)
 (3.2347067e+02 7.2707135e-03)(3.2350978e+02 7.6510271e-03)(3.2354888e+02 8.0600150e-03)
 (3.2358798e+02 8.5000576e-03)(3.2362708e+02 8.9736583e-03)(3.2366618e+02 9.4834142e-03)
 (3.2370528e+02 1.0031968e-02)(3.2374438e+02 1.0621938e-02)(3.2378348e+02 1.1255817e-02)
 (3.2382258e+02 1.1935834e-02)(3.2386168e+02 1.2663767e-02)(3.2390078e+02 1.3440702e-02)
 (3.2393988e+02 1.4266718e-02)(3.2397898e+02 1.5140513e-02)(3.2401808e+02 1.6058946e-02)
 (3.2405718e+02 1.7016533e-02)(3.2409629e+02 1.8004909e-02)(3.2413539e+02 1.9012324e-02)
 (3.2417449e+02 2.0023248e-02)(3.2421359e+02 2.1018201e-02)(3.2425269e+02 2.1973952e-02)
 (3.2429179e+02 2.2864192e-02)(3.2433089e+02 2.3660804e-02)(3.2436999e+02 2.4335697e-02)
 (3.2440909e+02 2.4863094e-02)(3.2444819e+02 2.5221984e-02)(3.2448729e+02 2.5398362e-02)
 (3.2452639e+02 2.5386819e-02)(3.2456549e+02 2.5191179e-02)(3.2460459e+02 2.4824024e-02)
 (3.2464370e+02 2.4305229e-02)(3.2468280e+02 2.3659806e-02)(3.2472190e+02 2.2915454e-02)
 (3.2476100e+02 2.2100249e-02)(3.2480010e+02 2.1240729e-02)(3.2483920e+02 2.0360552e-02)

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 (3.2511290e+02 1.4843489e-02)(3.2515200e+02 1.4228585e-02)(3.2519110e+02 1.3661390e-02)
 (3.2523021e+02 1.3140435e-02)(3.2526931e+02 1.2663778e-02)(3.2530841e+02 1.2229188e-02)
 (3.2534751e+02 1.1834295e-02)(3.2538661e+02 1.1476692e-02)(3.2542571e+02 1.1154007e-02)
 (3.2546481e+02 1.0863955e-02)(3.2550391e+02 1.0604369e-02)(3.2554301e+02 1.0373222e-02)
 (3.2558211e+02 1.0168637e-02)(3.2562121e+02 9.9888893e-03)(3.2566031e+02 9.8324064e-03)
 (3.2569941e+02 9.6977663e-03)(3.2573851e+02 9.5836900e-03)(3.2577761e+02 9.4890353e-03)
 (3.2581672e+02 9.4127890e-03)(3.2585582e+02 9.3540590e-03)(3.2589492e+02 9.3120664e-03)
 (3.2593402e+02 9.2861376e-03)(3.2597312e+02 9.2756976e-03)(3.2601222e+02 9.2802636e-03)
 (3.2605132e+02 9.2994392e-03)(3.2609042e+02 9.3329101e-03)(3.2612952e+02 9.3804399e-03)
 (3.2616862e+02 9.4418672e-03)(3.2620772e+02 9.5171026e-03)(3.2624682e+02 9.6061278e-03)
 (3.2628592e+02 9.7089936e-03)(3.2632502e+02 9.8258199e-03)(3.2636413e+02 9.9567954e-03)
 (3.2640323e+02 1.0102178e-02)(3.2644233e+02 1.0262295e-02)(3.2648143e+02 1.0437547e-02)
 (3.2652053e+02 1.0628406e-02)(3.2655963e+02 1.0835422e-02)(3.2659873e+02 1.1059222e-02)
 (3.2663783e+02 1.1300519e-02)(3.2667693e+02 1.1560111e-02)(3.2671603e+02 1.1838893e-02)
 (3.2675513e+02 1.2137855e-02)(3.2679423e+02 1.2458098e-02)(3.2683333e+02 1.2800836e-02)
 (3.2687243e+02 1.3167406e-02)(3.2691153e+02 1.3559281e-02)(3.2695064e+02 1.3978076e-02)
 (3.2698974e+02 1.4425568e-02)(3.2702884e+02 1.4903707e-02)(3.2706794e+02 1.5414630e-02)
 (3.2710704e+02 1.5960685e-02)(3.2714614e+02 1.6544447e-02)(3.2718524e+02 1.7168743e-02)
 (3.2722434e+02 1.7836681e-02)(3.2726344e+02 1.8551671e-02)(3.2730254e+02 1.9317460e-02)
 (3.2734164e+02 2.0138166e-02)(3.2738074e+02 2.1018318e-02)(3.2741984e+02 2.1962895e-02)
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 (3.2781085e+02 3.6475515e-02)(3.2784995e+02 3.8641061e-02)(3.2788905e+02 4.0986700e-02)
 (3.2792815e+02 4.3528632e-02)(3.2796725e+02 4.6284239e-02)(3.2800635e+02 4.9272003e-02)
 (3.2804545e+02 5.2511353e-02)(3.2808456e+02 5.6022402e-02)(3.2812366e+02 5.9825527e-02)
 (3.2816276e+02 6.3940752e-02)(3.2820186e+02 6.8386855e-02)(3.2824096e+02 7.3180120e-02)
 (3.2828006e+02 7.8332632e-02)(3.2831916e+02 8.3850015e-02)(3.2835826e+02 8.9728473e-02)
 (3.2839736e+02 9.5951046e-02)(3.2843646e+02 1.0248299e-01)(3.2847556e+02 1.0926626e-01)
 (3.2851466e+02 1.1621332e-01)(3.2855376e+02 1.2320048e-01)(3.2859286e+02 1.3006167e-01)
 (3.2863196e+02 1.3658361e-01)(3.2867107e+02 1.4250412e-01)(3.2871017e+02 1.4751554e-01)
 (3.2874927e+02 1.5127551e-01)(3.2878837e+02 1.5342689e-01)(3.2882747e+02 1.5362761e-01)
 (3.2886657e+02 1.5158854e-01)(3.2890567e+02 1.4711538e-01)(3.2894477e+02 1.4014686e-01)
 (3.2898387e+02 1.3078081e-01)(3.2902297e+02 1.1928007e-01)(3.2906207e+02 1.0605434e-01)
 (3.2910117e+02 9.1619792e-02)(3.2914027e+02 7.6544085e-02)(3.2917937e+02 6.1387834e-02)

(3.2921848e+02 4.6653611e-02)(3.2925758e+02 3.2750215e-02)(3.2929668e+02 1.9975180e-02)
 (3.2933578e+02 8.5140259e-03)(3.2937488e+02 -1.5480745e-03)(3.2941398e+02 -1.0207505e-02)
 (3.2945308e+02 -1.7521025e-02)(3.2949218e+02 -2.3585618e-02)(3.2953128e+02 -2.8521368e-02)
 (3.2957038e+02 -3.2458114e-02)(3.2960948e+02 -3.5525868e-02)(3.2964858e+02 -3.7848471e-02)
 (3.2968768e+02 -3.9539836e-02)(3.2972678e+02 -4.0702087e-02)(3.2976588e+02 -4.1425017e-02)
 (3.2980499e+02 -4.1786391e-02)(3.2984409e+02 -4.1852765e-02)(3.2988319e+02 -4.1680564e-02)
 (3.2992229e+02 -4.1317263e-02)(3.2996139e+02 -4.0802557e-02)(3.3000049e+02 -4.0169470e-02)
 (3.3003959e+02 -3.9445343e-02)(3.3007869e+02 -3.8652721e-02)(3.3011779e+02 -3.7810107e-02)
 (3.3015689e+02 -3.6932618e-02)(3.3019599e+02 -3.6032543e-02)(3.3023509e+02 -3.5119819e-02)
 (3.3027419e+02 -3.4202437e-02)(3.3031329e+02 -3.3286776e-02)(3.3035239e+02 -3.2377887e-02)
 (3.3039150e+02 -3.1479733e-02)(3.3043060e+02 -3.0595385e-02)(3.3046970e+02 -2.9727187e-02)
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(3.4704839e+02 -7.7360852e-04)(3.4708749e+02 -7.6226610e-04)(3.4712659e+02 -7.5117431e-04)
(3.4716569e+02 -7.4032557e-04)(3.4720479e+02 -7.2971256e-04)(3.4724389e+02 -7.1932825e-04)
(3.4728299e+02 -7.0916590e-04)(3.4732209e+02 -6.9921899e-04)(3.4736119e+02 -6.8948127e-04)
(3.4740029e+02 -6.7994672e-04)(3.4743939e+02 -6.7060952e-04)(3.4747849e+02 -6.6146409e-04)
(3.4751760e+02 -6.5250504e-04)(3.4755670e+02 -6.4372716e-04)(3.4759580e+02 -6.3512546e-04)
(3.4763490e+02 -6.2669510e-04)(3.4767400e+02 -6.1843141e-04)(3.4771310e+02 -6.1032990e-04)
(3.4775220e+02 -6.0238622e-04)(3.4779130e+02 -5.9459617e-04)(3.4783040e+02 -5.8695570e-04)
(3.4786950e+02 -5.7946089e-04)(3.4790860e+02 -5.7210796e-04)(3.4794770e+02 -5.6489324e-04)
(3.4798680e+02 -5.5781320e-04)(3.4802590e+02 -5.5086442e-04)(3.4806500e+02 -5.4404358e-04)
(3.4810411e+02 -5.3734748e-04)(3.4814321e+02 -5.3077302e-04)(3.4818231e+02 -5.2431719e-04)
(3.4822141e+02 -5.1797708e-04)(3.4826051e+02 -5.1174988e-04)(3.4829961e+02 -5.0563285e-04)
(3.4833871e+02 -4.9962335e-04)(3.4837781e+02 -4.9371882e-04)(3.4841691e+02 -4.8791677e-04)
(3.4845601e+02 -4.8221479e-04)(3.4849511e+02 -4.7661054e-04)(3.4853421e+02 -4.7110175e-04)
(3.4857331e+02 -4.6568622e-04)(3.4861241e+02 -4.6036181e-04)(3.4865152e+02 -4.5512646e-04)
(3.4869062e+02 -4.4997813e-04)(3.4872972e+02 -4.4491489e-04)(3.4876882e+02 -4.3993483e-04)
(3.4880792e+02 -4.3503610e-04)(3.4884702e+02 -4.3021691e-04)(3.4888612e+02 -4.2547552e-04)
(3.4892522e+02 -4.2081023e-04)(3.4896432e+02 -4.1621940e-04)(3.4900342e+02 -4.1170142e-04)
(3.4904252e+02 -4.0725474e-04)(3.4908162e+02 -4.0287784e-04)(3.4912072e+02 -3.9856925e-04)
(3.4915982e+02 -3.9432753e-04)(3.4919892e+02 -3.9015129e-04)(3.4923803e+02 -3.8603917e-04)
(3.4927713e+02 -3.8198983e-04)(3.4931623e+02 -3.7800201e-04)(3.4935533e+02 -3.7407443e-04)
(3.4939443e+02 -3.7020589e-04)(3.4943353e+02 -3.6639518e-04)(3.4947263e+02 -3.6264114e-04)
(3.4951173e+02 -3.5894266e-04)(3.4955083e+02 -3.5529862e-04)(3.4958993e+02 -3.5170795e-04)
(3.4962903e+02 -3.4816960e-04)(3.4966813e+02 -3.4468256e-04)(3.4970723e+02 -3.4124582e-04)
(3.4974633e+02 -3.3785842e-04)(3.4978543e+02 -3.3451941e-04)(3.4982454e+02 -3.3122786e-04)
(3.4986364e+02 -3.2798287e-04)(3.4990274e+02 -3.2478357e-04)(3.4994184e+02 -3.2162909e-04)
(3.4998094e+02 -3.1851859e-04)(3.5002004e+02 -3.1545126e-04)(3.5005914e+02 -3.1242629e-04)
(3.5009824e+02 -3.0944291e-04)(3.5013734e+02 -3.0650036e-04)(3.5017644e+02 -3.0359788e-04)
(3.5021554e+02 -3.0073475e-04)(3.5025464e+02 -2.9791027e-04)(3.5029374e+02 -2.9512373e-04)
(3.5033284e+02 -2.9237445e-04)(3.5037195e+02 -2.8966179e-04)(3.5041105e+02 -2.8698508e-04)
(3.5045015e+02 -2.8434369e-04)(3.5048925e+02 -2.8173700e-04)(3.5052835e+02 -2.7916441e-04)
(3.5056745e+02 -2.7662532e-04)(3.5060655e+02 -2.7411916e-04)(3.5064565e+02 -2.7164535e-04)
(3.5068475e+02 -2.6920335e-04)(3.5072385e+02 -2.6679261e-04)(3.5076295e+02 -2.6441260e-04)
(3.5080205e+02 -2.6206280e-04)(3.5084115e+02 -2.5974270e-04)(3.5088025e+02 -2.5745180e-04)

(3.5091935e+02 -2.5518962e-04)(3.5095846e+02 -2.5295569e-04)(3.5099756e+02 -2.5074952e-04)
 (3.5103666e+02 -2.4857068e-04)(3.5107576e+02 -2.4641870e-04)(3.5111486e+02 -2.4429315e-04)
 (3.5115396e+02 -2.4219360e-04)(3.5119306e+02 -2.4011964e-04)(3.5123216e+02 -2.3807084e-04)
 (3.5127126e+02 -2.3604681e-04)(3.5131036e+02 -2.3404715e-04)(3.5134946e+02 -2.3207147e-04)
 (3.5138856e+02 -2.3011939e-04)(3.5142766e+02 -2.2819055e-04)(3.5146676e+02 -2.2628457e-04)
 (3.5150587e+02 -2.2440109e-04)(3.5154497e+02 -2.2253978e-04)(3.5158407e+02 -2.2070028e-04)
 (3.5162317e+02 -2.1888225e-04)(3.5166227e+02 -2.1708538e-04)(3.5170137e+02 -2.1530932e-04)
 (3.5174047e+02 -2.1355377e-04)(3.5177957e+02 -2.1181841e-04)(3.5181867e+02 -2.1010294e-04)
 (3.5185777e+02 -2.0840705e-04)(3.5189687e+02 -2.0673046e-04)(3.5193597e+02 -2.0507287e-04)
 (3.5197507e+02 -2.0343401e-04)(3.5201417e+02 -2.0181358e-04)(3.5205327e+02 -2.0021133e-04)
 (3.5209238e+02 -1.9862698e-04)(3.5213148e+02 -1.9706027e-04)(3.5217058e+02 -1.9551094e-04)
 (3.5220968e+02 -1.9397874e-04)(3.5224878e+02 -1.9246342e-04)(3.5228788e+02 -1.9096473e-04)
 (3.5232698e+02 -1.8948244e-04)(3.5236608e+02 -1.8801631e-04)(3.5240518e+02 -1.8656611e-04)
 (3.5244428e+02 -1.8513162e-04)(3.5248338e+02 -1.8371260e-04)(3.5252248e+02 -1.8230884e-04)
 (3.5256158e+02 -1.8092013e-04)(3.5260068e+02 -1.7954626e-04)(3.5263978e+02 -1.7818701e-04)
 (3.5267889e+02 -1.7684218e-04)(3.5271799e+02 -1.7551158e-04)(3.5275709e+02 -1.7419500e-04)
 (3.5279619e+02 -1.7289226e-04)(3.5283529e+02 -1.7160316e-04)(3.5287439e+02 -1.7032751e-04)
 (3.5291349e+02 -1.6906513e-04)(3.5295259e+02 -1.6781585e-04)(3.5299169e+02 -1.6657948e-04)
 (3.5303079e+02 -1.6535585e-04)(3.5306989e+02 -1.6414478e-04)(3.5310899e+02 -1.6294612e-04)
 (3.5314809e+02 -1.6175969e-04)(3.5318719e+02 -1.6058532e-04)(3.5322630e+02 -1.5942287e-04)
 (3.5326540e+02 -1.5827217e-04)(3.5330450e+02 -1.5713307e-04)(3.5334360e+02 -1.5600541e-04)
 (3.5338270e+02 -1.5488905e-04)(3.5342180e+02 -1.5378383e-04)(3.5346090e+02 -1.5268962e-04)
 (3.5350000e+02 -1.5160626e-04)

A. 174

B. 274

C. 374

D. 474

E. 574

F. 674

G. 774

H. 874

I. 974

J. 1074

K. 1174

L. 1274

答案

正确答案: **F**

详细解析

该谱图为典型的零场分裂参数 $E = 0$ 的轴对称型双自由基冻结溶液谱图。

CHECKPOINT

1 PTS

零场分裂参数 $E = 0$ 的轴对称型双自由基

自旋间的平均间距 $\langle r \rangle$ 可以通过零场分裂参数 D 值估算，其与自旋间关系为 $D/mT = 2789/(\langle r/\text{Angstrom} \rangle^3)$ 。

CHECKPOINT

1 PTS

$$D/mT = 2789/(\langle r/\text{Angstrom} \rangle^3)$$

EPR谱两侧的峰对应z方向吸收，其距离等于两倍D值。读谱可知 $2D/mT \approx 18.18mT$ ，故 $D = 9.09mT$ 。

CHECKPOINT

1 PTS

$$D = 9.09mT$$

代入上述关系可求得 $\langle r \rangle = 6.74 \text{ Angstrom}$ ，对应于 674 pm。因此选F。

CHECKPOINT

1 PTS

$$\langle r \rangle = 6.74 \text{ Angstrom} = 674 \text{ pm}$$